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## **1. Introduction**

Predictive analysis refers to the usage of data to answer questions, identify trends to strategize and to make impactful decisions. Predictive analysis is often associated with the field of big data and computer science. Algorithms find specific patterns to extract insights from data and forecast future events (IBM, 2024). Regardless of the field, predictive analysis proves useful in providing insights to make decisions (Cote, 2024). Advancements in machine learning have helped expand predictive analytics capabilities. Popular predictive analysis models include classification and clustering. The former falls under supervised learning and the latter falls under unsupervised learning (IBM, 2024).

### **Supervised Learning**

Supervised learning is categorized under machine learning and artificial intelligence. It makes use of labelled datasets to train algorithms for accurate data classification (IBM, 2024).

#### **Classification models**

Classification models belong to the field of supervised machine learning. (IBM, 2024). Classification is the process of grouping the data into predefined categories based on their specific attributes (Simplilearn, 2024). An example of a use case of this model would be using it to classify prospects or customers into groups for segmentation. Another example of a use case can be to simply use it to answer yes or no questions for things like fraud detection (IBM, 2024).

### **Unsupervised Learning**

Unsupervised learning is a subcategory of machine learning and artificial intelligence that deals with unlabelled data (geeksforgeeks, 2024). Unsupervised learning algorithms have no previous knowledge regarding the data and are tasked with finding patterns and relationships between the data (geeksforgeeks, 2024).

#### **Clustering models**

Clustering models fall under the branch of unsupervised machine algorithms. Clustering involves grouping data based on similarities and dissimilarities into clusters (Simplilearn,

2024). An example of a use case of this model could be to separate customers into similar groups for marketing effectively to each group. Examples of clustering models are: k-means clustering, hierarchical clustering and mean-shift clustering.

This project explores the problem domain of Student dropout. Student dropout is a huge socioeconomic issue around the world. There can be various reasons as to why a student may choose to dropout and quit studies. These reasons may include financial issues, mental health issues or motivation or life changes. Issues students can have financially include cost of tuition, cost of additional expenses and cost of living (Mowreader, 2024). Issues that students may face related to their mental wellbeing/ mental health include Depression, Anxiety and Substance Abuse (Writers, 2024). All of these reasons have the ability to lead a student to dropping out.

Ensuring that students finish their education is known to be highly correlated with the future success of the student. Student dropout has the potential to negatively impact the lives of the students and the institutions where the students are enrolled (Realinho, et al., 2022). For students, it can limit their future employment opportunities and wellbeing. For educational institutions, high dropout rates can appear poorly on their records and affect their reputation. On a societal level, high dropout rates can hinder economic growth. Because this is such a big challenge, research regarding student dropout has been going on since as early as the 1970s (Tinto, 1975).

In this project, the focus is on applying predictive analysis techniques to predict student dropout. To achieve this, three supervised learning algorithms are going to be used.

All three algorithms that are going to be used fall under Supervised learning under the subcategory of Classification Models. An introduction to the models is provided below and they will be further explored in the [solution](#) section.

### **K-Nearest Neighbours (KNN):**

K-Nearest Neighbours (KNN) is a supervised learning algorithm that does not assume any specific data distribution. It makes classifications or predictions based on the proximity of data points. KNN is suitable for classification as well as for regression but is generally used as a classification algorithm (IBM, 2024).

**Naïve Bayes:**

The Naïve Bayes classifier is a type of supervised learning algorithm that operates based on Bayes' Theorem. Bayes' Theorem is used to calculate conditional probability (Chauhan, 2022). Conditional probability is the likelihood of a result occurring given a past outcome in similar circumstances (Hayes, 2024).

**Random Forest:**

Random forest is group of decision trees that cooperate and work together in order to provide a single output (geeksforgeeks, 2024).

By using these models, the project aims to provide a way to predict student dropouts with a goal of improving educational outcomes.

## 2. Background

### Similar Projects

In the last two decades, as digitalized data has become more available, it has become easier to analyse the factors that lead to students dropping out (Vaarma & Li, 2024). Higher education institutes record a significant amount of data regarding their students which makes a considerable potential to generate information, extract knowledge and perform monitoring (Realinho, et al., 2022). This has been possible through the usage of machine learning techniques.

Researchers have explored traditional machine learning algorithms such as Naïve Bayes, Logistic Regression, Decision Trees (DT), K-Nearest Neighbours, Random Forest and Support Vector Machines (SVM) to establish prediction models for the sake of student dropout prediction. Other researchers have also taken leverage of neural network methods such as CNN (Convolutional Neural Network) to predict student dropout (Chi, et al., 2023). Several notable studies and projects have focused on the area of Student Dropout Prediction. All of these make use of certain evaluation metrics to assess their models. The metrics used are accuracy, recall, precision, F1 score and AUC. To understand the results given by these metrics, understanding True Positives, True Negatives, False Positives and False Negatives is essential.

**True Positive:** A true positive occurs when the prediction was positive, and the actual result is positive as well (Sahil, 2020).

**True Negative:** A true negative occurs when the prediction was negative, and the actual result is negative as well (Sahil, 2020).

**False Positive:** A false positive occurs when the prediction was positive, but the actual result is negative (Sahil, 2020).

**False Negative:** A false negative occurs when the prediction was negative, but the actual result is positive (Sahil, 2020).

The metrics used are explained below:

**Accuracy:** Accuracy is a machine learning metric that tells us how often a model makes the right prediction compared to the total number of predictions it makes. Accuracy is measured using the formula given below:

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total Number of predictions}}$$

Figure 1: Accuracy Formula

**Recall:** Recall is a machine learning evaluation metric that calculates the ratio of actual positives that were correctly classified as positives by the model. Recall is measured using the formula given below:

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False negatives}}$$

Figure 2: Recall Formula

**Precision:** Precision is a machine learning evaluation metric that calculates the ratio of true positive predictions to the total number of positive predictions made by the model. It measures the accuracy of the positive predictions (Mari, et al., 2024). Precision is measured using the formula given below:

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False positives}}$$

Figure 3: Precision Formula

**F1 Score:** F1 score is a machine learning metric that takes into consideration both precision and recall. F1 score is measured using the formula given below:

$$\text{F1 score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

Figure 4: F1 Score Formula

**AUC:** AUC shows the likelihood that when a positive and negative example are randomly picked, the model will rank the positive example higher than the negative one (Google Developer Program, 2024). It demonstrates the model's capability to differentiate between positive and negative classes

The notable studies that apply these metrics to assess models for student dropout prediction are:

**Study 1: “Early Dropout Prediction in Online Learning of University using Machine Learning”** (Park & Yoo, 2021)

**Description:** This study focuses on predicting dropout rates among students of online universities in Korea. For more than a decade, data was collected in the learning management system (LMS) of Korea's online colleges which is utilized in this study (Park & Yoo, 2021). The study makes use of four different machine learning algorithms: Random Forest, Support Vector Machines (SVM), Deep Neural Network (DNN).

The performance of the models was judged using various evaluation metrics. These metrics include Accuracy, Recall, Precision, F1 Score, and AUC (Park & Yoo, 2021). The following are the results of the evaluation metrics:

| Study                                                                            | Model         | Accuracy | Recall | Precision | F1 Score | AUC  |
|----------------------------------------------------------------------------------|---------------|----------|--------|-----------|----------|------|
| Early Dropout Prediction in Online Learning of University using Machine Learning | Decision Tree | 0.91     | 0.70   | 0.66      | 0.68     | 0.82 |
|                                                                                  | Random Forest | 0.96     | 0.76   | 0.96      | 0.84     | 0.95 |
|                                                                                  | SVM           | 0.81     | 0.55   | 0.31      | 0.40     | 0.69 |
|                                                                                  | DNN           | 0.85     | 0.39   | 0.41      | 0.37     | 0.77 |

*Table 1: Evaluation metrics of study 1*

From these results, it is evident that Random Forest has the highest values in most of the evaluation metrics. At the end of the study, it was concluded that Random Forest was the best for predicting dropout of online learners.

## Study 2: “Towards Predicting Student’s Dropout in University Courses Using Different Machine Learning Techniques” (Kabathova & Drlik, 2021)

**Description:** This study investigates dropout prediction among students of the e-learning course of LMS Moodle. The study analyses the data collected from four academic years (Kabathova & Drlik, 2021). It makes use of six different machine learning algorithms: Naïve Bayes, Random Forest, Neural Network, Logistic Regression, Support Vector Machine and Decision Tree

The performance of the models was judged using various evaluation metrics. These metrics include Accuracy, Recall, Precision and F1 Score (Kabathova & Drlik, 2021). The following are the results of the evaluation metrics:

| Study                                                                                                  | Model                        | Accuracy | Recall | Precision | F1 Score |
|--------------------------------------------------------------------------------------------------------|------------------------------|----------|--------|-----------|----------|
| Towards Predicting Student’s Dropout in University Courses Using Different Machine Learning Techniques | Naïve Bayes                  | 0.77     | 0.72   | 0.93      | 0.82     |
|                                                                                                        | Random Forest                | 0.93     | 0.96   | 0.86      | 0.91     |
|                                                                                                        | Neural Network               | 0.88     | 0.89   | 0.86      | 0.88     |
|                                                                                                        | Logistic Regression          | 0.93     | 0.98   | 0.79      | 0.87     |
|                                                                                                        | Support Vector Machine (SVM) | 0.92     | 0.96   | 0.79      | 0.88     |
|                                                                                                        | Decision Tree (DT)           | 0.90     | 0.98   | 0.71      | 0.85     |

Table 2: Evaluation metrics of study 2

The three algorithms that evidently perform the best are Support Vector Machines, Logistic Regression, and Random Forest. Among them, Random Forest shows high recall and precision. Logistic Regression achieved the highest recall but has less precision. Support Vector Machines also performs well but has low precision. Hence, at the end of the study, it was concluded that the best dropout prediction was obtained by Random Forest (RF) classifier.

**Study 3: “Analysis and Prediction of MOOC Learners’ Dropout Behaviour”** (Chi, et al., 2023)

**Description:** This study analyses the data from a Massive Open Online Course (MOOC) to identify the factors that are associated with learners’ dropout behaviour. It employs machine learning algorithms logical regression, K Nearest Neighbour (KNN) and Random Forest for the dropout prediction (Chi, et al., 2023).

The performance of the models was judged using various evaluation metrics. These metrics include Accuracy, Recall, Precision, F1 Score, and AUC (Chi, et al., 2023). The following are the results of the evaluation metrics:

| Study                                                       | Model               | Accuracy | Recall | Precision | F1 Score | AUC  |
|-------------------------------------------------------------|---------------------|----------|--------|-----------|----------|------|
| Analysis and Prediction of MOOC Learners’ Dropout Behaviour | Random Forest       | 0.92     | 0.99   | 0.93      | 0.95     | 0.93 |
|                                                             | Logistic Regression | 0.91     | 0.98   | 0.93      | 0.95     | 0.91 |
|                                                             | KNN                 | 0.92     | 0.98   | 0.93      | 0.95     | 0.93 |

*Table 3: Evaluation metrics of study 3*

From these results, evidently, Random Forest achieved the best overall performance across metrics. It had almost a perfect recall of 0.99. Logistic Regression and KNN also performed well however Random Forest stood out from the three. At the end of the study, it was concluded that the prediction effect of Random forests was found to be better than that of KNN and logical regression model.

## Comparison of Results

| Study   | Best performing Model | Accuracy | Recall | Precision | F1 score | AUC       |
|---------|-----------------------|----------|--------|-----------|----------|-----------|
| Study 1 | Random Forest         | 0.96     | 0.76   | 0.96      | 0.84     | 0.95      |
| Study 2 | Random Forest         | 0.93     | 0.96   | 0.86      | 0.91     | Not Given |
| Study 3 | Random Forest         | 0.92     | 0.99   | 0.93      | 0.95     | 0.93      |

Table 4: Comparison of Results

Across all three studies, Random Forest has consistently shown the best performance in predicting student dropout. Other algorithms such as KNN, Logistic Regression and Support Vector Machines (SVM) also perform well but Random Forest has consistently outshined them.

## Dataset Description

The dataset that will be used in this project is sourced from the [UC Irvine Machine Learning Repository](#). The dataset includes 37 columns and 4424 instances. Each instance of the dataset represents a student. Short descriptions of all of the columns of this dataset are given below:

| Column Name                    | Column Description                                                                                         |
|--------------------------------|------------------------------------------------------------------------------------------------------------|
| Marital Status                 | Marital status: 1 - Single, 2 - Married, 3 - Widower, 4 - Divorced, 5 - Facto Union, 6 - Legally Separated |
| Application Mode               | Application mode with specific codes representing various types of admission                               |
| Application Order              | Application order (between 0 - first choice; and 9 last choice)                                            |
| Course                         | Course identifier with specific codes for various programs                                                 |
| Daytime/evening attendance     | 1 – daytime 0 – evening                                                                                    |
| Previous Qualification         | Previous qualification level (e.g., Secondary education, Higher education)                                 |
| Previous Qualification (grade) | Grade of previous qualification (between 0 and 200)                                                        |
| Nacionality                    | Nationality codes (e.g., 1 - Portuguese, 2 - German, etc.)                                                 |
| Mother's Qualification         | Mother's education level (e.g., Secondary education, Higher education)                                     |
| Father's Qualification         | Father's education level (e.g., Secondary education, Higher education)                                     |
| Mother's Occupation            | Mother's occupation (coded with various professional categories)                                           |
| Father's Occupation            | Father's occupation (coded with various professional categories)                                           |
| Admission Grade                | Admission grade (between 0 and 200)                                                                        |
| Displaced                      | 1 - Yes, 0 - No (Whether the student is displaced)                                                         |

|                                                                         |                                                                     |
|-------------------------------------------------------------------------|---------------------------------------------------------------------|
| Educational special needs                                               | 1 - Yes, 0 - No (Whether the student has special educational needs) |
| Debtor                                                                  | 1 - Yes, 0 - No (Whether the student owes fees)                     |
| Tuition fees up to date                                                 | 1 - Yes, 0 - No (Whether tuition fees are up-to-date)               |
| Gender                                                                  | 1 - Male, 0 - Female                                                |
| Scholarship holder                                                      | 1 - Yes, 0 - No (Whether the student is a scholarship holder)       |
| Age at enrolment                                                        | Age of the student at enrolment                                     |
| International                                                           | 1 - Yes, 0 - No (Whether the student is an international student)   |
| Curricular <a href="#">units</a> 1 <sup>st</sup> semester (credited)    | Number of curricular units credited in the 1st semester             |
| Curricular <a href="#">units</a> 1 <sup>st</sup> semester (enrolled)    | Number of curricular units enrolled in the 1st semester             |
| Curricular <a href="#">units</a> 1 <sup>st</sup> semester (evaluations) | Number of evaluations to curricular units in the 1st semester       |
| Curricular <a href="#">units</a> 1 <sup>st</sup> semester (approved)    | Number of curricular units approved in the 1st semester             |
| Curricular <a href="#">units</a> 1st semester (grade)                   | Grade average in the 1st semester (between 0 and 20)                |
| Curricular <a href="#">units</a> 1st semester (without evaluations)     | Number of curricular units without evaluations in the 1st semester  |
| Curricular <a href="#">units</a> 2nd semester (credited)                | Number of curricular units credited in the 2nd semester             |
| Curricular <a href="#">units</a> 2nd semester (enrolled)                | Number of curricular units enrolled in the 2nd semester             |

|                                                                     |                                                                                                                                                                             |
|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Curricular <a href="#">units</a> 2nd semester (evaluations)         | Number of evaluations to curricular units in the 2nd semester                                                                                                               |
| Curricular <a href="#">units</a> 2nd semester (approved)            | Number of curricular units approved in the 2nd semester                                                                                                                     |
| Curricular <a href="#">units</a> 2nd semester (grade)               | Grade average in the 2nd semester (between 0 and 20)                                                                                                                        |
| Curricular <a href="#">units</a> 2nd semester (without evaluations) | Number of curricular units without evaluations in the 1st semester                                                                                                          |
| Unemployment rate                                                   | Unemployment rate (%)                                                                                                                                                       |
| Inflation rate                                                      | Inflation rate (%)                                                                                                                                                          |
| GDP                                                                 | GDP                                                                                                                                                                         |
| Target                                                              | Target. The problem is formulated as a <a href="#">three-category</a> classification task (dropout, enrolled, and graduate) at the end of the normal duration of the course |

Table 5: Dataset Description

The dataset covers demographic variables such as Marital status and Nationality. Marital status may cause variations in financial stability and time constraints due to which dropout and graduation rates may differ between married and unmarried students. Different nationalities may face language barriers or cultural integration challenges which can make certain nationality groups experience higher dropout rates compared to others.

Academic and socioeconomic variables such as previous qualifications, parental income are also covered by the dataset. Students with better achievements or prior degrees may show lower dropout rates and higher graduation rates. And variables such as Mother's or Father's occupation may show certain occupation groups experiencing higher dropout rates compared to others.

The target variable is divided into three categories: "dropout", "enrolled" and "graduate" which reflect the main outcomes of student enrolment.

By analysing all of these features, institutions can identify what factors have the most influence in leading students to dropping out and they can tailor their decisions accordingly.

### 3. Solution

The goal of this project is to combat high Student Dropout rates by building a model that makes predictions about which students are at risk of dropping out. The predicting will be done based on demographic factors such as Marital status and Nationality, academic and socioeconomic factors such as previous qualifications, parental income as well as other factors provided by the dataset. To achieve this, three popular basic machine learning algorithms will be used: Naive Bayes, K-Nearest Neighbours (KNN), and Random Forest.

These three algorithms will assist in analysing student data and identifying trends that can forecast whether a student will remain enrolled or dropout. Each of these algorithms work differently and will produce different results. In the end, by comparing the algorithms through various assessment metrics, the most suitable algorithm for this particular problem will be identified. This approach to the solution provides a final reliable model for predicting student dropouts.

Before diving into the algorithms to be used, it is essential to understand these terms:

**Overfitting:** Overfitting is a behavior in which a machine learning model performs good on training data but it does not perform well on any new data (aws, 2024).

**Underfitting:** Underfitting is a behaviour in which a machine learning model is too simple to grasp the complexities of the data (geeksforgeeks, 2024) .

The proposed algorithms for the prediction model are as follows:

#### Naïve Bayes

The Naïve Bayes classifier falls under supervised learning. It is based on Bayes' Theorem. Bayes' Theorem is used to calculate conditional probability (Chauhan, 2022). Conditional probability likelihood of a result occurring given a past outcome in similar circumstances (Hayes, 2024).

Formula for Bayes Theorem:

$$P(Y|X) = \frac{P(X|Y) \cdot P(Y)}{P(X)}$$

Where:

$P(Y|X)$  represents the probability of  $Y$  (the label or outcome) given we have  $X$  (the input features)

$P(X|Y)$  represents the probability of observing the features ( $X$ ) if we already know the outcome ( $Y$ )

$P(Y)$  represents the overall likelihood of  $Y$  occurring

$P(X)$  represents the overall likelihood of observed features  $X$  across all classes.

Figure 5: Bayes' Theorem

The Naïve Bayes' classifier is called "Naïve" bayes because of the assumptions that it makes. The Naïve Bayes' classifier makes two assumptions. One is that each feature makes an independent contribution to the outcome. And the other is that every feature contributes equally to the outcome (Chauhan, 2022). The types of Naïve Bayes' classifier are:

### **Gaussian Naïve Bayes'**

Works with numerical (continuous) data and it assumes that numerical features follow a Gaussian (normal) distribution

### **Multinomial Naïve Bayes'**

Works with discrete data Used for discrete data like text classification and it assumes that features represent the frequency of events.

### **Bernoulli Naïve Bayes'**

Works with binary features and it assumes that each feature is binary (either 0 or 1)

Because of the naïve assumption made by the Naïve Bayes' classifier, this algorithm is very fast and efficient. This is because it requires less time to train and predict in comparison to most complex models. However, this classifier assumes that all of the features are independent which is not a realistic scenario for many datasets.

**Pseudocode of Naïve Bayes:**

**SET UP** the Naïve Bayes' classifier

**FIT** the model to the training dataset.

**ITERATE** over each class  $C$  in the target variable:

**CALCULATE**  $P(C)$

**ITERATE** over every feature  $X$

**CALCULATE**  $P(X|C)$  using the Gaussian formula

**PREDICT** labels for the test data

**ITERATE** over each test example

**CALCULATE**  $P(C|X) = P(X|C) * P(C)$

**ASSIGN** the class with the highest  $P(C|X)$  as the predicted label.

**EVALUATE** the model's performance

**OUTPUT** the evaluation results

**The flowchart for Naïve Bayes is given in the next page**

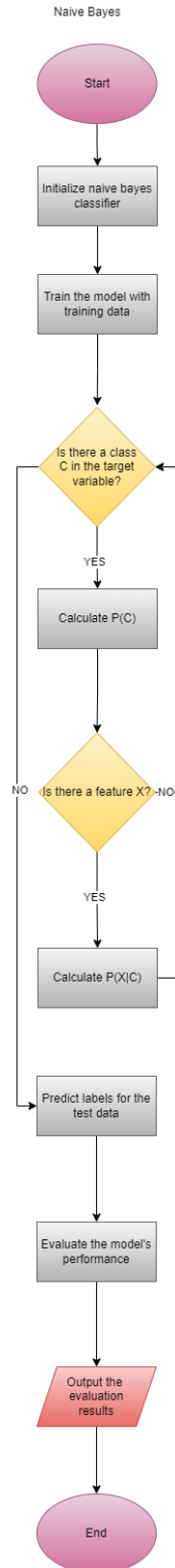


Figure 6: Naive Bayes Flowchart

## KNN

K-Nearest Neighbours (KNN) is a supervised learning algorithm that does not assume any specific data distribution. It makes classifications or predictions based on the proximity of data points. KNN can be used for classification as well as regression. It is typically found to be used for classification algorithms. This is how KNN works for classification:

- 1) When a new data point comes in, KNN calculates the distance of the data point from all stored points using the Euclidean distance formula:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + \dots + (z_2 - z_1)^2}$$

Where:

$x_1, x_2, y_1, y_2, \dots, z_1, z_2$  are the coordinates of two points

$d$  is the Euclidean distance between the two points.

*Figure 7: Euclidean distance*

- 2) KNN finds the  $k$  closest points which are called “neighbours” to the data. “ $k$ ” refers to the number of neighbours. While choosing the value of “ $k$ ” the following factors should be considered:

- A small  $k$  value can be too specific leading to overfitting.
- A large  $k$  value can be too generalized leading to underfitting

To find the best value of  $k$ , the following things can be done:

- Cross Validation: Testing different values of  $k$  to find out which one works the best
- Elbow Method: Plotting the error rate against different values of  $k$ . As the value of  $k$  is increased, the error rate decreases initially but then it flattens out or starts to increase slightly. The “elbow” point of this curve is considered the optimal value of  $k$ . This is where the error rate is low and increasing  $k$  further does not provide any benefits.

- 3) The data point is assigned with the most common label among these members.

**Pseudocode of KNN:**

**SET UP** the KNN classifier with a specified value for k.

**FIT** the KNN model to the training dataset.

**ITERATE** over each test example

**CALCULATE** distance using the Euclidean distance formula

**IDENTIFY** the k nearest neighbours based on distance

**CONDUCT** majority voting in order to identify the predicted class.

**EVALUATE** the performance of the model

**OUTPUT** the evaluation results

**The flowchart of KNN is given in the next page**

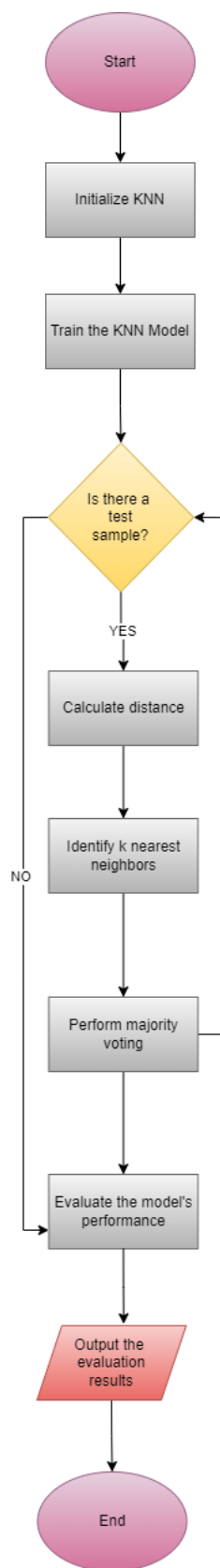


Figure 8: KNN Flowchart

## Random Forest

To understand random forest, understanding decision trees first is essential.

### Decision Trees

Decision Tree is a supervised parameter that is nonparametric. It consists of a root node, branches, internal nodes and leaf nodes (IBM, 2024). It can be used for classification tasks as well as regression tasks (IBM, 2024). A decision tree begins from a root node. From the root node flows a series of nodes called decision nodes. Each decision node represents a split point. Then from the decision nodes stem leaf nodes. The leaf nodes of the decision tree represent the possible answers (Master's in Data Science, 2024). Here is a diagram which represents a typical decision tree:

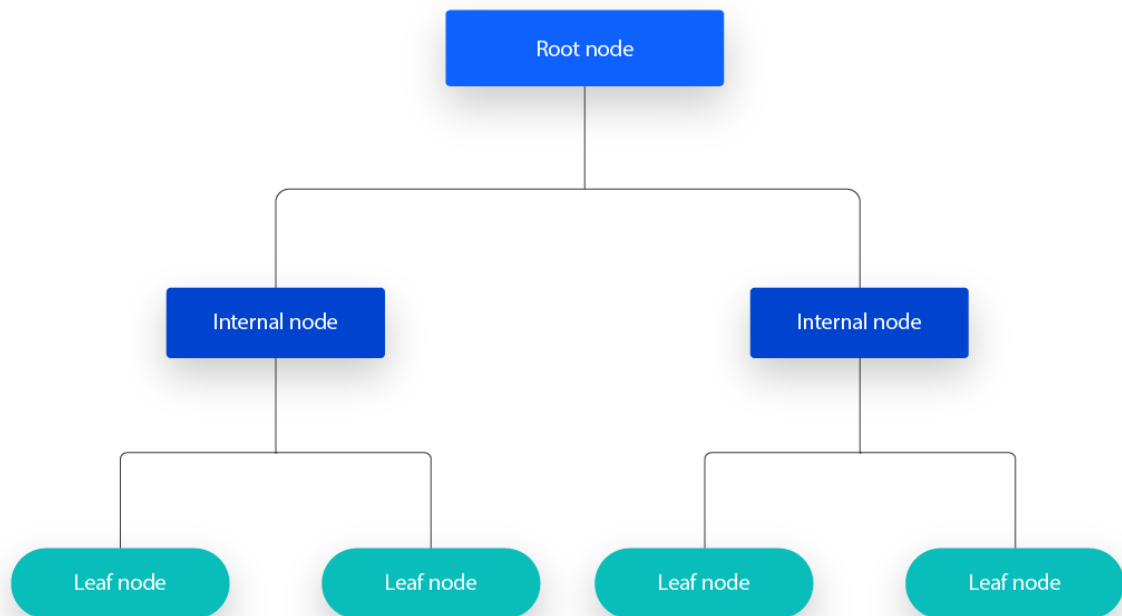


Figure 9: Decision Tree (IBM, 2024)

Random Forest falls under ensemble learning models. Ensemble learning consists of models, either of the same type or different, which team up to enhance predictive

performance (geeksforgeeks, 2024). Random forest works by generating a number of trees during the training phase. Each tree in the random forest is created using a random subset of the data. Randomness allows variability amongst all of the trees which prevents overfitting. (geeksforgeeks, 2024). Variability amongst individual trees also improves the prediction performance as a whole. For prediction, the algorithm aggregates the result of all trees. The way it combines the predictions is different for classification problems and regression problems. For classification problems, the results from all the trees are combined by majority voting. The class that gets the most votes in the end is selected as the final prediction.

On the other hand, for regression problems, the results from all the trees are combined by averaging. This process leads to the bias of the trees being smoothed out and gives a more accurate result.

In this project, random forest is used for classification. Hence, the algorithm will aggregate the results of all trees by voting. Here is a diagram which represents a random forest:

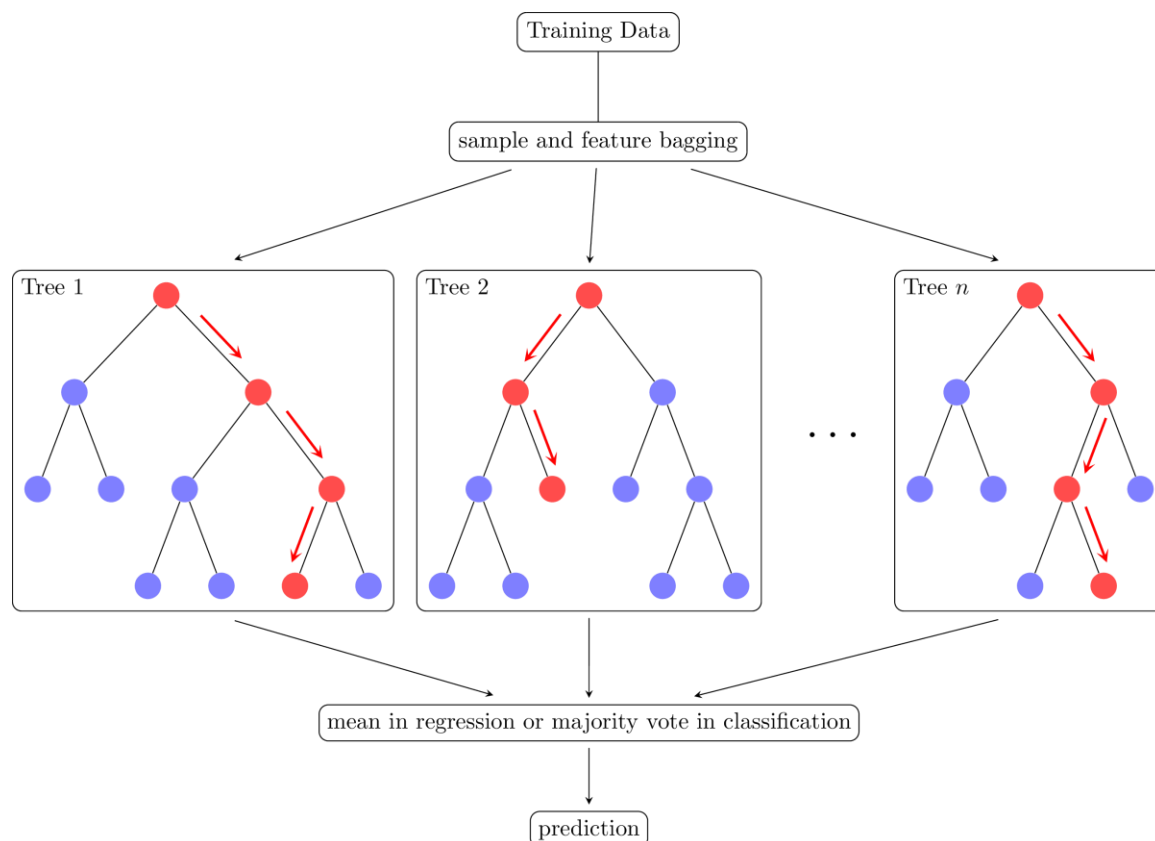


Figure 10: Random Forest (Riebesell, 2024)

**Pseudocode of Random Forest:**

**SET UP** the Random Forest classifier with T trees

**FIT** the Random Forest model to the training data

**ITERATE** over every tree in the forest

**SELECT** a random subset of training data

**SELECT** a random subset of features

**BUILD** the decision tree

**PREDICT** using all trees

**ITERATE** over each tree

**OUTPUT** the predicted class for the test sample

**COMBINE** predictions using majority voting

**COUNT** how many times each class appears in all the trees

**SET** the class that appears most frequently as the predicted label.

**EVALUATE** performance of the model

**OUTPUT** evaluation results

**The Flowchart of Random Forest is given in the next page**

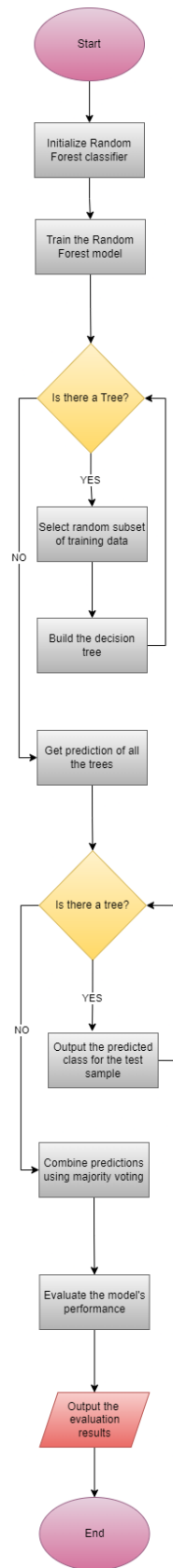


Figure 11: Random Forest Flowchart

**Why these algorithms were chosen:**

| Algorithm     | Reason for choosing                                                                                                                                                                                                     |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| KNN           | KNN was selected because the dataset to be used for this project consists of numerical features that can effectively use proximity-based measures for classification.                                                   |
| Naïve Bayes   | Naïve Bayes assumes feature independence which simplifies calculations and makes it computationally efficient.                                                                                                          |
| Random Forest | Random forest is an ensemble model which combines the power of multiple decision trees (geeksforgeeks, 2024). Its resistance to overfitting and its ability to handle complex relationships makes it a reliable choice. |

*Table 6: Reasons for choosing the algorithms*

## 4. Conclusion

In this proposal, a machine learning model is presented for predicting student dropout which is a big challenge for all educational institutions. The goal is to provide higher education institutions with a tool that detects at risk students early. This allows the institutions to perform timely interventions. Through the analysis of student data such as academic achievement, attendance and socioeconomic characteristics, the system will enable institutions to focus their resources on the areas where it is the most needed. The focus of resources on these areas will ultimately lead to improved student retention.

The proposed system will use three algorithms: Naïve Bayes, K-Nearest Neighbours (KNN) and Random Forest. Each of these algorithms provide unique advantages which makes this combined approach even better. Naïve Bayes simplifies calculations and is computationally efficient. KNN uses proximity-based measure which is suitable for this project as it consists of numerical features. Random Forest is resistant to overfitting and is reliable at handling complex relationships.

By evaluating the performance of each of these algorithms, the model which provides the most accurate predictions for student dropout will be determined. During the course of the project, the performance of each algorithm will be assessed using various evaluation metrics. These metrics include accuracy, precision, recall, and F1 score. This will help identify the final model that gives the best predictions for Student Dropout.

However, as with any model, there are limitations and areas for improvement. For this student dropout prediction model, these include:

1. The accuracy of this model depends on availability of reliable and comprehensive student data. This might not always be realistic.
2. Although Random Forest and KNN are generally efficient, they can be a bit intensive, particularly with large datasets, and may require runtime and memory optimization.
3. The ensemble structure of Random Forest might make it more difficult to grasp, which may be an issue for stakeholders that want clear explanations of model predictions.

This system can still be enhanced through the inclusion of additional features such as: mental health status, student engagement with extracurricular activities which can help

improve predictions. As the project progresses, it might also be beneficial to explore alternative algorithms such as Support Vector Machines (SVM) and Gradient Boosting in order to improve predictions and maximize performance.

In conclusion, the proposed machine learning model that makes use of Naive Bayes, KNN, and Random Forests offers a workable and efficient way to forecast student dropout. This has the potential to greatly increase retention rates, which would ultimately lead to better educational results by assisting educational institutions in early identification of at-risk students.

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